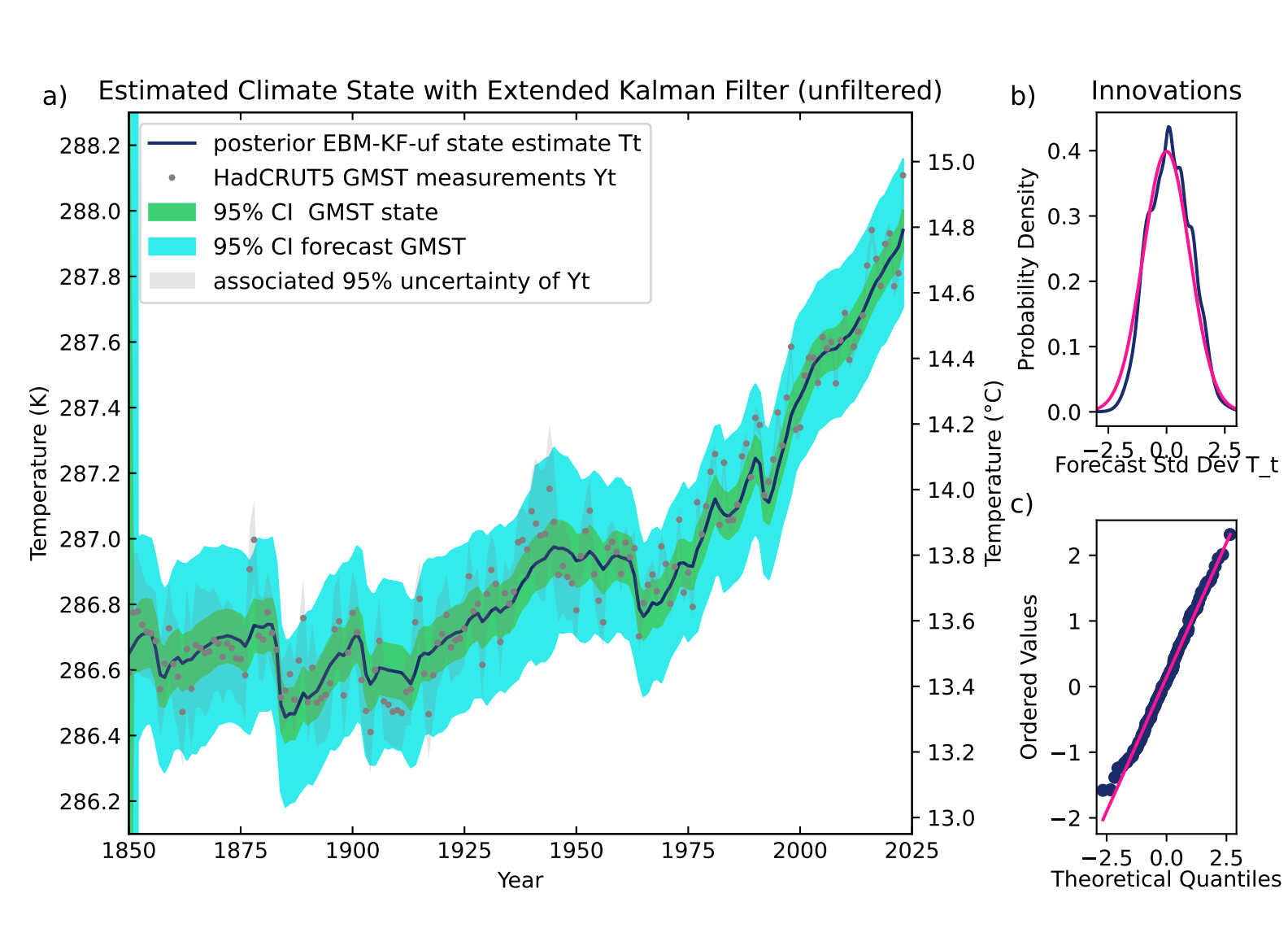
a) Estimated Climate State with Extended Kalman Filter (unfiltered)
posterior EBM-KF-uf state estimate Tt
HadCRUT5 GMST measurements Yt
95% CI GMST state
95% CI forecast GMST
associated 95% uncertainty of Yt
Temperature (K)
Temperature (°C)
Year
288.2
288.0
287.8
287.6
287.4
287.2
287.0
286.8
286.6
286.4
286.2
15.0
14.8
14.6
14.4
14.2
14.0
13.8
13.6
13.4
13.2
13.0
1850
1875
1900
1925
1950
1975
2000
2025
b) Innovations
Probability Density
0.4
0.3
0.2
0.1
0.0
−2.5
0.0
2.5
Forecast Std Dev T_t
c)
Ordered Values
2
1
0
−1
−2
−2.5
0.0
2.5
Theoretical Quantiles